

Qatar-2b

R-1331

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_210418 · created 2026-07-31T20:36:14+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2459322.7324 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.2 +/- 0.015
Transit depth (Rp/Rs)^2	4.02 +/- 0.62 [%]
Orbital Inclination (inc)	85.14 +/- 1.64
Airmass coefficient 1 (a1)	0.963 +/- 0.017
Airmass coefficient 2 (a2)	0.0236 +/- 0.0084
Scatter in the residuals of the lightcurve fit is	1.75 %
Best Comparison Star	#2 - [100, 49]
Optimal Method	PSF photometry
Transit Duration (day)	0.0696 +/- 0.007

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.2 ± 0.015 vs 0.16327 (deviation 2.45σ)
Duration fit vs published	0.0696 d vs 0.0767 d (deviation 1.01σ)
Mid-time O-C	1.4 min
Depth SNR	13.3
Residual scatter	1.75 %

Light curve

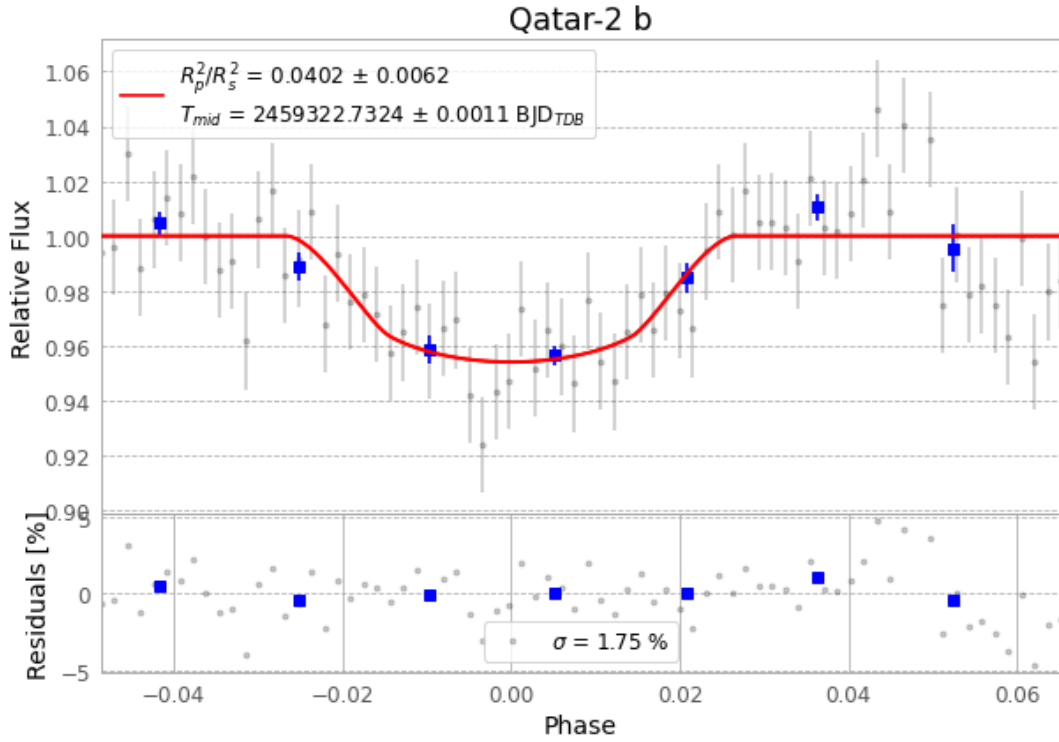


Figure 1 — FinalLightCurve_Qatar-2 b_2021-04-17.png (SHA-256 99434cd19119b5e8...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [529, 60], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 65482b8f80f0961d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

74 FITS frames (49 MB), Qatar-2210418035112.FITS → Qatar-2210418073012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-210418.log (7a852da38463...), FinalLightCurve_Qatar-2 b_2021-04-17.png (99434cd19119...), FinalLightCurve_Qatar-2 b_2021-04-17.csv (a0791eb06a87...)

Compute resources

Wall time 120.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 20:36Z.